QuantRIC: A Hybrid Quantum-Classical Framework for Predictive ISAC-RIS Orchestration in 6G O-RAN

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***Abstract*—The convergence of Integrated Sensing and Communication (ISAC) and Reconfigurable Intelligent Surfaces (RIS) is a key enabler for Sixth Generation (6G) networks, facilitating adaptive propagation for environment-aware operation. However, conventional approaches rely on reactive optimization driven by instantaneous Channel State Information (CSI), which suffers from rapid aging under mobility and blockages. Furthermore, the high-dimensional and non-convex nature of these optimizations imposes significant computational overhead and poor scalability in dynamic 6G environments. Such limitations lead to inconsistent RIS configurations and degraded joint sensing–communication performance. This paper introduces *QuantRIC*, a hybrid quantum–classical framework for predictive ISAC–RIS orchestration. It utilizes a multimodal transformer model to fuse radio and visual inputs into a unified latent representation of the 6G scenario. From this latent state, the model extracts task-specific descriptors for sensing and communication, which serve as inputs to a dual-branch Variational Quantum Circuit (VQC). This VQC jointly optimizes both objectives to generate a single RIS phase configuration within a bounded computational complexity. Simulation results shows *QuantRIC* sensing mean squared error (MSE) of 0.0855 line of sight (LoS) and 0.0332 non line of sight (NLoS), while the communication MSE remains at 0.0377 LoS and 0.0397 NLoS.**

***Index Terms*—Integrated Sensing and Communications, Radio Access Network Intelligent Controller, Reconfigurable Intelligent Surfaces, Variational Quantum Circuits.**

I. Introduction

The evolution toward Sixth Generation (6G) wireless systems is driven by applications such as autonomous transportation and industrial automation that demands both reliable communication and timely awareness of the physical environments [1], [2]. This dual requirement has led to the emergence of Integrated Sensing and Communication (ISAC) mechanism which supports joint sensing and data transmission within the same wireless infrastructure [3]. However, the practical realization of ISAC places stringent demands on the Open Radio Access Network (O-RAN) architecture of 6G networks, responsible for management of higher-level functions such as network control and data handling [4]. To manage such complex systems, the RAN Intelligent Controller (RIC) acts as the brain of the system. It serves as the primary orchestration entity where decisions are made to dynamically optimize resource allocation based on fluctuating user and environmental conditions [5], [6].

Despite these architectural advances in the RIC, the shared utilization of frequency and time resources for ISAC creates a fundamental performance trade-off between sensing accuracy and communication throughput. This challenge is compounded by the large scale mobility and physical obstructions of 6G environment that may lead to unpredictable signal degradation [7], [8]. Although the RIC can optimize transmission strategy, it remains a passive observer of the physical channel, unable to rectify deep fades or blockages once the signal has left the antenna [9]. To address this gap in end-to-end channel controllability, the Reflecting Intelligent Surface (RIS) has emerged as a transformative solution to convert the wireless channel from a passive, disruptive medium into an active, software-defined network component [10], [11].

When integrated into the O-RAN architecture, the RIS configurations are computed by the RIC and disseminated as control commands for dynamic adjustments to the reflective properties of the surface elements [12]. This formulation yields a non-convex optimization problem due to the unit-modulus constraints of the RIS phase shifts and the complex coupling of ISAC objectives. To circumvent the resulting mathematical intractability, various frameworks have emerged to approximate optimal performance. These existing ISAC-RIS solutions are broadly categorized into signal processing and learning based approaches [13], [14].

The former approach improve sensing accuracy through Direction-of-Arrival (DOA) estimation and sparse reconstruction techniques, but remain largely decoupled from network-level control and do not support closed-loop RIS orchestration in dynamic environments [15]. On the other hand, the later approach employs Deep Learning (DL) and Large Language Models (LLMs) to solve joint localization and communication objectives [16]–[19]. Despite their adaptability, these methods

*Code is available at: https://github.com/Mahdiya-Nishat/QuantRIC.

rely on high-dimensional classical neural networks, resulting in substantial parameter complexity, slow convergence, and poor scalability with increase in RIS elements. As a result, they fail to achieve the performance-to-complexity balance required for large-scale 6G deployments, motivating the need for a fundamentally different ISAC-RIS orchestration paradigm.

Thus, this paper investigates the following research questions (RQs):

- **RQ1:** How can system resilience be maintained in the presence of stochastic signal blockages and channel failures?
- **RQ2:** How can large-scale RIS arrays be managed to prevent the computational bottlenecks inherent in dynamic 6G optimization environments?
- **RQ3:** How can both sensing and communication be optimized be optimized simultaneously while ensuring long-term system stability?

To address the aforementioned challenges, this work proposes a optimization control framework that integrates multimodal sensing with a bounded-complexity variational solver to enable proactive RIS orchestration and system stability under high mobility.

The main contributions of this paper are summarized as follows:

- We designed a multimodal transformer model that fused radio-frequency signals with visual scene information to mitigate channel impairment and partial observability through robust environment modeling.
- We develop a scalable RIS control mechanism that transforms high-dimensional phase optimization into a low-dimensional latent representation, enabling efficient phase synthesis with bounded computational complexity for large-scale deployments.
- We propose a dual-branch variational quantum circuit (VQC) framework that jointly optimizes sensing and communication to ensure balanced performance and stable operation in dynamic 6G environments.

The remainder of this paper is organized as follows. Section II presents the system model and the formalization of the joint ISAC-RIS optimization problem. Section III details the *QuantRIC* framework including the multimodal fusion logic and the design of the VQC solver. Section IV provides an performance evaluation and Section V concludes the paper with a discussion on future research directions.

II. System Model and Problem Formulation

This section describes the system model of an RIS-assisted ISAC architecture operating in a smart traffic scenario, followed by the formulation of the joint sensing and communication control problem.

A. System Model

We consider a 6G enabled smart city traffic environment, where vehicles move through an urban environment with frequent signal blockage and reflections. To maintain reliable sensing and communication, a RIS is deployed and actively controlled by the RIC at the O-RAN to shape dynamic signal propagation as illustrated in Fig. 1. To prevent inaccurate RIS adjustments in such dynamic conditions, we capture two synchronized streams of data from the same environment, consisting of radio observations and visual information.

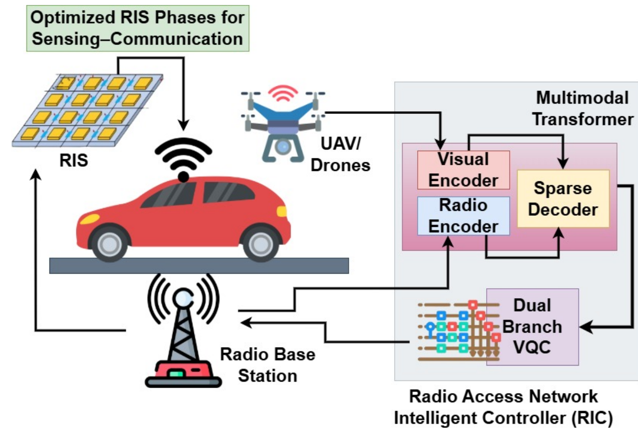


Fig. 1: The radio and visual observations processed through modality-specific encoders, are fused via a sparse cross-attention mechanism within the decoder. This yields two learnable query tokens, dedicated to sensing and communication that serves as inputs to a dual-branch VQC to generate the optimal RIS phase configurations for dynamic 6G scenario.

The radio signals reflected from vehicles $D_R(t)$ are received at the base station antennas and visual data $D_V(t)$ is captured by drones and roadside cameras, to provide a complementary views of the scene. These multimodal inputs are processed together using a transformer-based model designed for joint representation learning. model. Specifically, we employ two parallel encoders extract features from each input streams and the outputs are passed to a unified decoder using cross attention mechanism. This process yields a unified latent representation z_t of the environment, expressed as:

$$z_t = f_W(D_R(t), D_V(t)) \,, \tag{1}$$

where $f_W(\cdot)$ is used in a short-horizon *look-ahead* mode: the inferred latent state is treated as a proxy of the environment over the next control interval (e.g., $t{+}1$), which reduces the impact of CSI aging on decisions.

Subsequently, two learnable query tokens, attend to the z_t from Eq. 1 to extract task-relevant information to yield sensing ($O_S(t)$) and communication ($O_C(t)$) oriented descriptors as:

$$g(z_t) = \big[O_S(t),\, O_C(t)\big]. \tag{2}$$

The descriptors $\big(O_S(t), O_C(t)\big)$ are then are passed through a linear projection and scaled to $[-\pi, \pi]$ via a hyperbolic tangent activation, to serve as the rotation angles for the VQC layer. The VQC is employed as a bounded-complexity optimization module that efficiently resolves the coupled sensing–communication trade-off and generates feasible RIS control parameters ϕ_t. These phase configurations are subsequently dispatched to the RIS reflective elements via the RIC to dynamically shape the wireless propagation environment to jointly optimize sensing accuracy and communication quality for each t. The feasibility of the system is ensured by decoupling training and execution, where model parameters are learned offline and only a single forward inference is performed at the RIC module of the O-RAN.

B. Problem Formulation

We consider an RIS composed of N reflective elements, controlled by a phase vector $\boldsymbol{\phi}_t \in \mathbb{C}^N$. Given the sensing descriptor $O_S(t)$, the sensing-oriented control problem seeks an RIS configuration that maximizes sensing utility:

$$\max_{\boldsymbol{\phi}_t} U_S(\boldsymbol{\phi}_t, O_S(t)) \quad \text{s.t.} \quad |\phi_{t,n}| = 1, \ \forall n. \tag{3}$$

Here, $U_S(\cdot)$ quantifies sensing performance through beampattern alignment.

Similarly, given the communication descriptor $O_C(t)$, the communication-oriented control problem maximizes communication utility under the same physical constraints:

$$\max_{\boldsymbol{\phi}_t} U_C(\boldsymbol{\phi}_t, O_C(t)) \quad \text{s.t.} \quad |\phi_{t,n}| = 1, \ \forall n. \tag{4}$$

Although defined separately, the objectives in Eq.(3) and Eq.(4) are intrinsically coupled, as both depend on the same RIS configuration $\boldsymbol{\phi}_t$. A configuration that favors sensing performance may therefore degrade communication throughput, and vice versa.

We thus formulate a joint optimization problem that balances both objectives:

$$\max_{\boldsymbol{\phi}_t} \ \alpha_t \, U_S(\boldsymbol{\phi}_t, O_S(t)) + (1-\alpha_t) \, U_C(\boldsymbol{\phi}_t, O_C(t)), \tag{5}$$

where $\alpha_t \in [0,1]$ controls the sensing–communication trade-off according to application-level priorities. This formulation motivates a low-latency solver capable of balancing sensing and communication objectives under predictive uncertainty. The following section presents the proposed framework that realizes this objective.

III. *QuantRIC*: The Proposed Framework

To address the joint sensing–communication objective in Eq.(5), the *QuantRIC* employs a dual-branch VQC comprising of $2k$ number of qubits. As illustrated in Fig. 2, the circuit operates on a quantum register partitioned into two disjoint branches defined in Eq. 2. The proposed VQC is organized into four successive layers to maximize circuit expressibility while maintaining bounded computational complexity for RIS phase control.

In Layer 1, each descriptor is angle-encoded using parameterized single-qubit rotations, mapping classical features into quantum states. The sensing descriptor $O_S(t)$ is encoded on the first k qubits, while the communication descriptor $O_C(t)$ is encoded on the remaining k qubits. In Layer 2, independent variational transformations are applied within each branch to capture intra-task correlations.

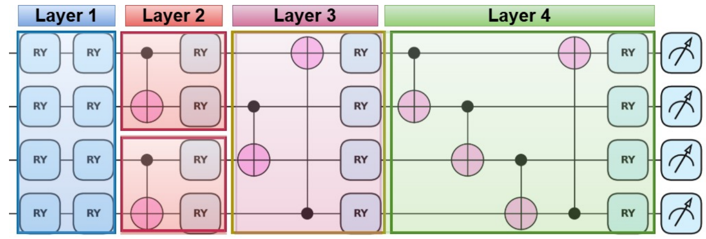


Fig. 2: Dual-branch VQC design: Sensing and communication descriptors are angle-encoded on disjoint qubit subsets, processed independently to capture intra-task correlations, and softly coupled through controlled entanglement to generate a unified RIS phase configuration.

In Layer 3, controlled entanglement is introduced between corresponding sensing and communication qubits to enable coordinated optimization under the shared RIS constraint. This design allows the VQC to balance competing objectives while maintaining a fixed circuit depth compatible with RIC operation. Finally, the Layer 4 applies a fixed global mixing pattern across the full register to improve circuit expressivity and improve feature mixing across the full register, without increasing circuit depth.

The VQC outputs a vector of Pauli expectation values $\mathbf{y}_t = \{\langle \psi_t \mid P_l \mid \psi_t \rangle\}_{l=1}^{2k} \in [-1,1]^{2k}$. These values are shifted and scaled to obtain the RIS phase configuration $\boldsymbol{\phi}_t = \pi(\mathbf{y}_t + 1) \in [0, 2\pi]^{2k}$. The Algorithm 1 details the proposed procedure. The resulting ϕ_t configuration are dispatched by the RIC to the RIS controller, which applies them across all reflecting elements to enable continuous adaptation to dynamic 6G channel conditions.

IV. Experimental Validation

We evaluate *QuantRIC* using the DeepMIMO dataset, specifically the O1_28B scenario, which models a standardized outdoor mmWave environment at 28 GHz with 12 base stations and 497,931 user locations. All experiments were conducted on a consumer-grade system equipped with an Intel Core i5 processor, 16 GB RAM, and an NVIDIA GeForce GTX 1650 GPU. The classical components were implemented in Python 3.11 using PyTorch with CUDA acceleration, while quantum circuit simulations were executed on a CPU backend using PennyLane (v0.44.1). Pretrained vision models were obtained from the HuggingFace Transformers library.

A. Multimodal Data Construction

The dataset was loaded directly from its raw `.mat` files using `scipy.io` without any intermediate preprocessing pipeline, extracting per-user path loss, three-dimensional location coordinates and LoS indicators to construct the radio feature vector $\mathbf{d}_r \in \mathbb{R}^7$, serving as input to the radio encoder. Since the DeepMIMO O1_28B scenario does not provide paired visual data, we construct a synthetic visual modality by generating a 224×224 RGB image for each user location.

Each image encodes the local propagation environment

Algorithm 1: *QuantRIC*: The Proposed dual branch VQC for RIS Phase Optimization

Input: z_t
Output: ϕ_t

for $t = 0, 1, 2, \dots, T$ **do**
- ▷ *Task descriptor separation*
- $(O_S(t), O_C(t)) \leftarrow g(z_t)$
- ▷ *Quantum register initialization*
- Initialize $2k$ qubits in $|0\rangle^{\otimes 2k}$
- ▷ ***Layer 1**: Angle encoding on disjoint branches*
- **for** $i = 0$ **to** $2k-1$ **do**
  - $$\alpha_i(t) \leftarrow \begin{cases} O_{S,i}(t), & i < k \\ O_{C,i-k}(t), & i \geq k \end{cases}$$
  - Apply $R_Z(\alpha_i(t))\, R_Y(\theta_i)$ on qubit i
- ▷ ***Layer 2**: Local entanglement within each branch*
- **for** $i = 0$ **to** $k-2$ **do**
  - CNOT$(i, i+1)$
  - CNOT$(i+k, i+k+1)$
- ▷ ***Layer 3**: Cross-branch coupling (sensing $\leftrightarrow$ communication)*
- **for** $i = 0$ **to** $k-1$ **do**
  - CNOT$(i, i+k)$
- ▷ ***Layer 4**: Global mixing (ring coupling within the full register)*
- **for** $i = 0$ **to** $2k-2$ **do**
  - CNOT$(i, i+1)$
- CNOT$(2k-1, 0)$
- ▷ *Variational state preparation*
- Prepare $|\psi_t\rangle = |\psi_t(\boldsymbol{\theta}, O_S(t), O_C(t))\rangle$
- ▷ *Pauli expectation measurement (fixed output dimension)*
- $y_t \leftarrow \{\langle\psi_t|P_l|\psi_t\rangle\}_{l=1}^{2k}$
- ▷ *Phase scaling to RIS control space*
- $\boldsymbol{\phi}_t \leftarrow \pi(\mathbf{y}_t + 1)$

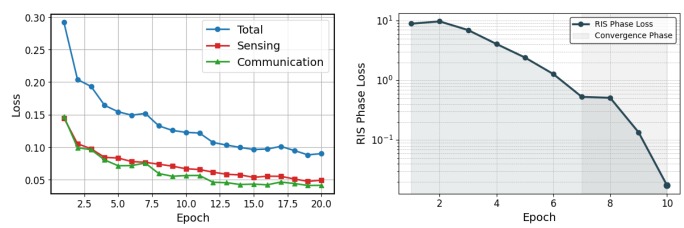
through a signal strength heatmap centered at the user position, location-dependent obstacle patterns, and a color-coded indicator for LoS and NLoS conditions. This representation serves as a proxy for drone captured imagery to provide spatial context to the visual encoder. Moreover, the images are deterministically generated from the underlying radio parameters, to ensure semantic alignment between the radio and visual modalities for effective multimodal fusion.

B. Model Configuration

The radio encoder is implemented as a 2 layer transformer with 4 attention heads and a feed-forward dimension of 1024, mapping the seven-dimensional radio feature vector to a 256-dimensional latent sequence. The visual encoder employs a pretrained ViT-Base/16 model with a 768-dimensional hidden space, whose output is projected to 256 dimensions to align with the radio representation. The fusion module and task decoder operate with four attention heads and a 256-dimensional hidden space, with the decoder utilizing two learnable query tokens for sensing and communication. Lastly, the dual-branch VQC is implemented in PennyLane using the `default.qubit` simulator, consisting of four qubits and four circuit layers, repeated four times to produce 16 RIS phase outputs. The complete set of simulation parameters is summarized in Table I. We next analyze these metrics through the results reported in the following figures.

TABLE I: Simulation parameters used for evaluating the *QuantRIC* framework.

Parameter	Value
Training samples	200
Evaluation samples	50
Radio feature dimension	7
Hidden dimension	256
Number of qubits	4
VQC circuit layers	4
VQC runs (RIS elements)	4 × 4 = 16
Learning rate	1×10^{-5}
Classical training epochs	20
VQC training epochs	10
VQC batch size	4
VQC learning rate	1×10^{-2}
Gradient clipping	1.0



(a) Stable and consistent reduction in total, sensing, and communication losses of the multimodal model.

(b) VQC loss curve on a logarithmic scale, illustrating rapid convergence over multiple orders of magnitude.

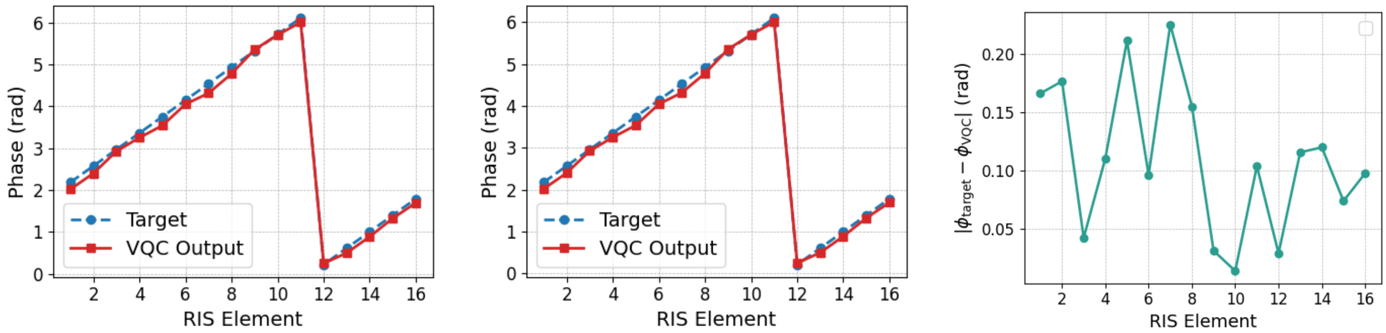
Fig. 3: Training dynamics of the *QuantRIC* framework

C. Training Convergence Analysis

We first analyze the training convergence behavior of the proposed *QuantRIC* framework. The Fig. 3a shows the evolution of the total, sensing, and communication losses of the multimodal model. All three components exhibit a consistent and stable reduction across epochs, indicating effective joint learning without oscillatory behavior or task imbalance. In particular, the sensing and communication losses converge smoothly while maintaining close alignment, demonstrating that the model captures shared structure across modalities without degrading individual task performance.

Figure 3b illustrates the convergence of the VQC-based RIS phase optimization on a logarithmic scale. After a brief initialization phase, the loss decreases rapidly over multiple orders of magnitude and stabilizes within a few epochs, indicating efficient optimization under bounded circuit depth. The shaded convergence region highlights the regime where the VQC reaches stable operation.

Together, these results demonstrate that the multimodal encoder and the quantum control module exhibit stable and



(a) Sample 1

(b) Sample 2

(c) Absolute Phase Error

Fig. 4: Comparison results show close alignment between predicted and target RIS phase shifts generated by the VQC, along with consistently low reconstruction error, indicating accurate and stable phase learning.

complementary convergence behavior, ensuring reliable end-to-end training of the proposed framework.

D. RIS Phase Configuration Analysis

Building on the observed stable convergence, we evaluate whether the learned representations yield accurate RIS phase configurations. Fig. 4a and Fig. 4b compare target and VQC-generated phase shifts across RIS elements for two samples. The predicted phases closely match the targets, showing consistent alignment across elements with smooth, monotonic progression, indicating effective capture of the spatial structure required for beamforming.

To further quantify this behavior, Fig. 4c shows the absolute phase error across two RIS elements. The error remains consistently low across all elements, with only minor variations, confirming that the VQC accurately approximates the desired phase configuration. A sharp transition observed in the phase plots corresponds to phase wrapping at 2π, which does not indicate a modeling error since such discontinuities represent physically equivalent configurations under the unit-modulus constraint. Despite this periodic behavior, the relative phase relationships are preserved, ensuring correct propagation control.

Thus, the results demonstrate that the proposed *QuantRIC* framework achieves accurate and stable phase reconstruction with strong generalization across samples.

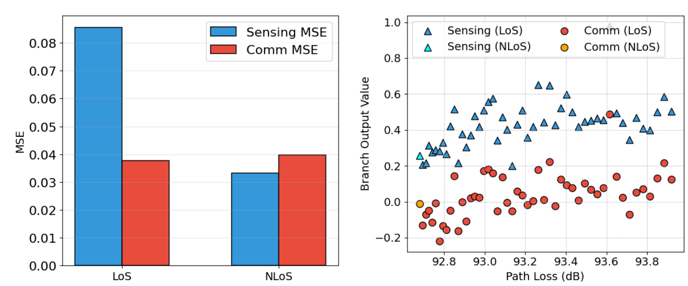
E. LoS and NLoS Performance Analysis

We next evaluate the behavior of the sensing and communication branches under varying channel conditions. The Fig. 5a illustrates the response of both branches as a function of path loss across 50 evaluation samples. The sensing branch consistently produces higher output magnitudes relative to the communication branch, which reflects its dependency on geometric information such as user location encoded in the radio features. Notably, the sensing output exhibits relatively stable values across the path loss range, suggesting that the sensing query token has learned to extract location-relevant features that are less sensitive to signal attenuation.

In contrast, the communication branch output shows lower magnitude with increased variability, indicating that the communication query token is more sensitive to propagation quality, physically consistent with the fact that beamforming gain is directly influenced by path loss.

Fig. 5b further quantifies MSE under LoS and NLoS conditions. The sensing branch shows a larger performance gap, as localization depends on direct paths and angle information that degrade under blockage. In contrast, the communication branch maintains relatively stable MSE across both conditions, indicating that multimodal fusion provides sufficient context to partially compensate for reduced channel quality in NLoS scenarios.

Together, these observations highlight the ability of the proposed framework to maintain balanced sensing and communication performance under varying channel environments.



(a) The sensing branch exhibits more stable output values, while the communication branch shows greater variability.

(b) The trade-off between sensing accuracy and communication performance across different propagation environments.

Fig. 5: Robustness and sensitivity analysis of *QuantRIC* under mobility and CSI uncertainty.

F. Addressal of Research Questions

RQ1: System resilience under stochastic blockages and CSI aging is addressed through the predictive multimodal formulation in Section II-A, which infers the environment state as a look-ahead latent representation (Eq. 1). Stable convergence in Fig. 3a validates this design, while Fig. 4 confirms robust phase reconstruction despite channel variability. Collectively, these results demonstrate that resilience is achieved by pivoting from reactive CSI-driven optimization to predictive multimodal state estimation.

RQ2: The scalability challenge of RIS control is addressed by the bounded-complexity VQC design (Section III), which reduces high-dimensional phase synthesis to fixed-depth quantum inference and linear scaling (Eq. 5). By replacing iterative solvers with a single forward pass, complexity is constrained by qubit count rather than RIS elements. Rapid convergence of the VQC loss in Fig. 3b validates this tractability, maintaining solution quality across scales.

RQ3: Joint sensing–communication optimization is achieved via the shared variational parameterization in Section III, coupling both objectives into a single RIS configuration (Eq. 5). Implemented through a dual-branch VQC, task-specific descriptors (Eq. 2) are encoded on disjoint qubit subsets and coordinated via cross-branch entanglement. This unified control space avoids conflicting updates by generating a single configuration ϕ_t per instant. Validation shows stable, distinct branch responses to path loss (Fig. 5a) and bounded MSE across LoS/NLoS conditions (Fig. 5b), maintaining a stable trade-off under dynamic 6G propagation.

V. Conclusion and Future Directions

This paper presents *QuantRIC*, a hybrid quantum–classical framework for predictive ISAC–RIS orchestration in 6G networks. By leveraging multimodal radio vision sensing, it enables look-ahead RIS control via latent environment representations, mitigating CSI aging and dynamic blockages. A dual-branch VQC jointly optimizes sensing and communication objectives within a unified parameterization, yielding optimal RIS configurations with bounded complexity.

Simulation results show stable convergence, with multimodal and VQC losses of 0.09 and 0.017, respectively. The framework preserves a strong sensing–communication trade-off, maintaining sensing magnitudes of 0.2–0.9 and balanced MSEs across LoS/NLoS conditions (Sensing: 0.086/0.033; Communication: 0.038/0.040), demonstrating effective objective decoupling with minimal error under channel variability.

Future work will extend *QuantRIC* to cooperative multi-RIS scenarios using distributed variational circuits for coordinated reflections, along with adaptive trade-off control under heterogeneous sensing priorities and hardware-in-the-loop validation on O-RAN testbeds.

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